

Unsupervised Learning - Lab 05 - Hierarchical Clustering

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Agenda

- Motivation:
 - This fifth lab session aims to apply hierarchical clustering algorithm and its variants to cluster an unknown matrix of data (with low dimensionality and continuous attributes).
- Steps:
 - Load the input data and import useful packages **[TASK 1]**
 - Prepare the dataset **[TASK 2]**
 - Design the clustering algorithm **[TASK 3]**
 - Apply the clustering algorithm to the dataset **[TASK 4]**
 - Use the clustering solution to form clusters **[TASK 5]**
 - Compute and visualize the cluster centers **[TASK 6]**
 - Validation **[TASK 7]**
 - Compute the inter-cluster, intra-cluster distances and the silhouette score

Instructions

- You are going to re-use already known packages (matplotlib, scipy, numpy, seaborn, scikit-learn.preprocessing). Check the previous lab solutions.
- Moreover, the `scipy.cluster.hierarchy` package will be introduced to easily cluster data using hierarchical clustering (see documentation [here](#)).
- Read the step-by-step instructions below carefully and write your own code to fill the missing steps in the Colab notebook (instructions are also reported in the notebook).

Useful Packages

- Useful packages: numpy, pandas, scipy, matplotlib, seaborn, sklearn, `scipy.cluster` (NEW!)
- Useful Python data structures: 2D matrix, list, ndarray

This time you will not be given the parameter K , then you need to guess an appropriate number of clusters by analysis.